

# Directed Reachability

Directed Graphs, BFS Reachability · Mentor-Led Lab (Student Edition)

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**Difficulty** ★★ Intermediate

## Problem

In a directed graph, follow the edges from a source vertex to find everything reachable. Report the BFS visiting order and the number of reachable vertices.

Edges are one-way, so the traversal must respect their direction.

## What to Compute

- **BFS Order:** the order vertices are first visited from the source.
- **Reachable:** the number of vertices reachable from the source (including itself).

## Input / Output Format

**Input:** Line 1: N M. Next M lines: a directed edge u v. Then a line with the source S.

**Output:** Two lines: BFS Order and Reachable.

## Examples

### Example 1

**Input:** 6 5 0 1 0 2 1 3 2 3 3 4 0

**Output:**

BFS Order: 0 1 2 3 4

Reachable: 5

**Explanation:** From 0 the directed edges reach 1, 2, 3 and 4; vertex 5 has no incoming path, so five vertices are reachable.

### Example 2

**Input:** 5 4 0 1 0 2 1 3 1 4 0

**Output:**

BFS Order: 0 1 2 3 4

Reachable: 5

**Explanation:** From 0 every other vertex is reachable along the directed edges.

## Constraints

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- Follow directed edges only in their direction.
- BFS explores neighbours in sorted order.
- Count the source among the reachable vertices.

## Sample Run

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### Sample Input

```
6 5
0 1
0 2
1 3
2 3
3 4
0
```

### Sample Output

```
BFS Order: 0 1 2 3 4
Reachable: 5
```